ROGERS CORPORATION

\\\ Datasheet

RO4450F™ and RO4460G2™ Bondply

The RO4400™ bondply family is comprised of several grades based on the RO4000 series core materials, and are compatible in multi-layer constructions with RO4000 laminates. A high postcure Tg makes RO4400 series bondply an excellent choice for multi-layers requiring sequential laminations as fully cured RO4400 bondplys are capable of handling multiple lamination cycles. In addition, FR-4 compatible bond requirements permit RO4400 bondply and low flow FR-4 bondply to be combined into non-homogeneous multi-layer constructions using a single bond cycle.

RO4450F™ bondply has demonstrated improvement in lateral flow capability and is becoming the first choice for new designs or as a replacement in designs that have difficult fill requirements. RO4460G2™ bondply provides designers with a 6.15 Dk bonding layer that, just as the other RO4400 bonding materials, exhibits excellent Dk control while maintaining a low z axis expansion for plated through hole reliability. Each of the bondplys are recognized by Underwriter Laboratories with the UL 94 V-0 flame rating, and are compatible with lead-free processes.



\\\ Features and Benefits:

Prepreg grades based on RO4000 series core materials

- Compatible in mulitlayer board constructions with RO4003C, RO4350B, RO4835, RO4350G2, and RO4000 LoPro laminates

Low z-axis coefficient of thermal expansion ranging from 43 to 60 ppm/°C

- CAF resistant

Sequential lamination capable

- High frequency thermoset prepreg compatible with FR-4 bond temperatures

Lead free solder processing compatible

- High reliability plated through-hole

\\\ Typical Applications:

- Backhaul Radios
- Communications Systems
- Power Amplifiers
- Small Cells/DAS

Standard Thicknesses	Standard Panel Sizes
0.0040" (0.102 mm) +/- 0.0007" *Contact Customer Service or Sales Engineering to inquire about additional available product configurations	16" X 18" (406 X 457mm) 24" X 18" (610 X 457mm) 24.5" X 18.5" (622 X 470mm) 24" X 36" (610 X 915 mm) *Additional panel sizes available

Properties	RO4450F	RO4460G2	Direction	Units	Test Conditions		Test Method
Electrical Properties							
Dielectric Constant	3.52 ± 0.05	6.15 ± 0.15	Z	-	23˚C @ 50% RH	10 GHz	IPC TM-650 2.5.5.5
Dissipation Factor	0.004	0.004	Z	-	23˚C @ 50% RH	10 GHz	IPC TM-650 2.5.5.5
Volume Resistivity	8.93 X10^{8}	9.1 X10^{8}	-	Mohm-cm	C-96/35/90	-	IPC TM-650 2.5.17.1
Surface Resistivity	1.03 X10^{7}	1.53 X10^{8}	X,Y	Mohm	C-96/35/90	-	IPC TM-650 2.5.17.1
Electrical Strength (dielectric strength)	1000	1000	Z	V/mil	D-48/50	-	IPC TM-650 2.5.6.2
Thermal Properties							
Decomposition Temperature (Td)	390	405	-	˚C	2hrs @ 105˚C	5% Weight Loss	IPC TM-650 2.3.40
Glass Transition (Tg)	>280	170	-	˚C TMA	A	-	IPC-TM-650 2.4.24.5
Coefficient of Thermal Expansion	19	15	X	ppm/˚C	-	-55˚C to 288˚C	IPC TM-650 2.4.41
Coefficient of Thermal Expansion	17	18	Y	ppm/˚C	-	-55˚C to 288˚C	IPC TM-650 2.4.41
Coefficient of Thermal Expansion	50	43	Z	ppm/˚C	-	-55˚C to 288˚C	IPC TM-650 2.4.41
Thermal Conductivity	0.65	0.67	Z	W/(m·K)	80˚C	z direction	ASTM D5470
Mechanical Properties							
Copper Peel Strength after Thermal Stress	0.70 (4.0)	1.04 (5.0)	Z	N/mm (lbs/in)	10s @288˚C	35 µm foil	IPC TM-650 2.4.8
Physical Properties							
Glass	1080	1080	-	-	-	-	-
Resin-Content	74	79	-	-	-	-	-
Color	White	White	-	-	-	-	-
Flammability	V-0	V-0	-	-	-	C48/23/50 & C168/70	UL 94
Moisture Absorption	0.04	0.05	-	%	D24/23	-	IPC TM-650 2.6.2.1
Lead Free Process Compatible	Yes	Yes	-	-	-	-	-

[1] Typical values are a representation of an average value for the population of the property. For specification values contact Rogers Corp.

\\\ 100 S. Roosevelt Avenue \\\ Chandler, AZ 85226 \\\ Tel: 480-961-1382 \\\ Fax: 480-961-4533 \\\ www.rogerscorp.com
\\\ IPC Slash Sheet # 4103B/540 \\\ UL File # E102763

The information in this data sheet is intended to assist you in designing with Rogers' circuit materials. It is not intended to and does not create any warranties express or implied, including any warranty of merchantability or fitness for a particular purpose or that the results shown on this data sheet will be achieved by a user for a particular purpose. The user should determine the suitability of Rogers' circuit materials for each application. These commodities, technology and software are exported from the United States in accordance with the Export Administration regulations. Diversion contrary to U.S. law prohibited.

The Rogers' logo, Helping power, protect, connect our world, RO4000, RO4003C, RO4350B, RO4835, RO4360G2, RO4835T, and LoPro are trademarks of Rogers Corporation or one of its subsidiaries.

©2022 Rogers Corporation, Printed in U.S.A., All rights reserved.
Issued 1595 090826 **Publication #92-150**

Helping power, protect, connect our world™

ROGERS CORPORATION